The three remaining Mathieu stabilizer labels

Human Compression project: L054 finite component

This note certifies rows 2, 5 and 7 of the retained current M_{11} companion table: M_{10}, $M_9 : 2$ and M_9. The printed table is dated 25 November 2012; identity with every historical web edition is not assumed. The accepted numerical row bijection and literal subgroup arrays are inputs. The notation is not inferred from subgroup orders.

Exact naming convention. GAP's reference manual, Section 50.1–13, provides the named Mathieu constructors in degrees nine and ten. The actual implementation in `grp/basicprm.gi` explicitly identifies M_{10} as the point-11 stabilizer in its standard M_{11}, and M_9 as the point-10 stabilizer in that M_{10}. We retain this named-constructor convention, including its literal generators; we do not claim a new characterization of Mathieu groups. The standard M_{11} is exactly the already acquired model

$$G = \langle (1\,2\,\cdots\,11),\ (3\,7\,11\,8)(4\,10\,5\,6) \rangle.$$

Its complete 7,920-element array is acquired from the accepted shared registry. Embedded on these eleven points, the retained generators are

$$\begin{aligned} M_{10}: &\quad (1\,9\,6\,7\,5)(2\,10\,3\,8\,4), \quad (1\,10\,7\,8)(2\,9\,4\,6), \\ M_9: &\quad (1\,4\,9\,8)(2\,5\,3\,6), \quad (1\,6\,5\,2)(3\,7\,9\,8). \end{aligned}$$

The verifier computes their closures from literal permutations, without using the constructor's stored orders. These closures have 720 and 72 elements and equal, element for element, G_{11} and $G_{10,11}$ obtained by filtering the acquired whole array. Thus the stabilizer joins are checked directly, in addition to retaining the names.

Theorem 1. *The actual subgroups matched to source rows 2, 7 and 5 are, respectively, a conjugate of this M_{10}, a conjugate of this M_9, and a split extension of this M_9 by a cyclic group of order two, with nontrivial complement action.*

Proof. All permutations use the GAP right-action convention. For each occurrence, the certificate supplies a literal $p \in G$. The verifier checks every element of $p^{-1}M_{10}p$ or $p^{-1}M_9p$ against the full closure of the corresponding saved subgroup generators. Conjugation is an isomorphism, so equality of these actual element sets proves the named identification. It additionally checks that row 2 is exactly G_9 and row 7 is exactly $G_{7,11}$.

For row 5 the full actual subgroup H equals $G_{\{7,11\}}$, the setwise pair stabilizer. Its pointwise kernel $N = G_{7,11}$ is the transported M_9 and has index two. The explicit element

$$t = (3\,10)(5\,9)(6\,8)(7\,11)$$

belongs to $H \setminus N$ and has order two. The verifier checks normalization, $H = N \sqcup Nt$, and a noncommuting pair from N and t. Consequently $H = N \rtimes \langle t \rangle$ with nontrivial action, exactly the displayed colon assertion. Neither a nonsplit extension nor a direct product is silently substituted for it. □

Certificate boundary. The row joins compare the original generator arrays and the exact native labels and punctuation, via the accepted numerical bijection. Three source joins require 864 full conjugacy-element checks and 72 additional normalization checks for the complement. Nine damaged certificates are rejected in ordinary and optimized Python. No GAP call, isomorphism search, ambient subgroup census or source-count filter occurs. The two new shared models have registry identifiers 34 and 35; every earlier registry remains unchanged. The author's root colleague suggested the stabilizer route and the reuse of saved elements, as disclosed in the packet.

Combined with the separately accepted 24-row pilot, corrected 17-row extension and four standard atoms, this accounts for all nine A_5 labels and all 39 M_{11} labels in these retained current tables. It does not certify structural labels in other tables, impose uniqueness on extension notation, or complete the whole source atlas.